SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1_AT3 — VISUALIZATION DESIGN EXERCISE

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO1 – Precision (BL2)	**Assessment:**	Assessment Tool 3 – Visualization Design
Weightage:	40% of CIA	**Total Marks:**	20 (2 Questions × 10 Marks)
CO1 Statement:	*Understand the fundamental concepts, principles, and techniques of data visualization.(BL2)*		
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Section / Batch:		**Date:**	28.07.2026

Code of Conduct

I Priyanka. G / 192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___________________________

Instructions

- This assessment contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Annexure A – VISUALIZATION DESIGN

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Problem 10 (10 Marks)

Scenario: A meteorologist wants to show wind direction (0°–360°) and wind speed (numeric) recorded every hour for one day.

Chart Type	Polar scatter / wind-rose style line-and-point plot
Coordinate System	Polar (r, θ)
Aesthetic Mapping	θ = wind direction, r = wind speed, color = hour of day
Scale Type	Circular/cyclical angular scale (θ) + continuous linear radial scale (r)
Color Scale	Sequential/cyclical scale for time-of-day (light → dark)

a) Design a chart that uses polar coordinates to represent both wind direction and speed simultaneously. Describe the mapping.

Design & Justification:

The proposed design is a polar scatter (wind-rose style) chart built on a polar coordinate system, since the dataset combines a circular, compass-based variable (direction) with a continuous magnitude variable (speed). Each hourly observation is plotted as a single point, and the two independent variables are mapped as follows:

- Angle (θ): Wind Direction (0°–360°): the compass direction from which the wind blows is mapped directly onto the angular position around the center of the plot, with 0°/360° oriented at the top (North) exactly as on a compass, so the geometry of the chart mirrors the real-world geometry of the variable it represents.
- Radius (r): Wind Speed (km/h): the recorded wind speed is mapped onto the distance of the point from the center, using a linear radial scale with concentric gridlines (e.g., 5, 10, 15, 20 km/h) so the viewer can read off an approximate speed for any point by comparing it against the gridlines.
- Connecting line: consecutive hourly points are joined with a thin line in chronological order, so the viewer can trace how direction and speed evolve together over the 24-hour period, revealing patterns such as a diurnal shift in wind direction.

This mapping allows both variables to be read from a single point: its angular position instantly communicates direction, while its distance from the center communicates speed something a single Cartesian chart cannot do as intuitively for a circular variable.

b) Why would Cartesian coordinates be a poor natural fit for the "direction" variable in this dataset?

Design & Justification:

Wind direction is a cyclical (circular) variable: it is measured in degrees from 0° to 360°, but 0° and 360° represent the exact same physical direction (due North), and the values wrap around continuously rather than having a true minimum or maximum.

A Cartesian (x, y) axis, by contrast, is inherently linear it assumes that values increase steadily along a straight line from a fixed low end to a fixed high end. If wind direction were plotted on a linear Cartesian x-axis, a wind that shifted gradually from 350° to 10° (a small, 20° change in reality) would appear on the chart as a dramatic jump from near the top of the axis down to near the bottom, creating an artificial and misleading discontinuity. Polar coordinates avoid this problem entirely because the angular axis is naturally circular, so 359° and 1° sit right next to each other on the plot exactly as they do in reality, making polar coordinates the structurally correct choice for this variable.

c) What additional aesthetic (e.g., color) could you add to also represent time of day in the same chart?

Design & Justification:

A color aesthetic can be layered onto the same points to encode the hour of the day (0–23) without needing a third spatial axis. Since hour-of-day is itself a cyclical, ordered numeric variable (hour 23 is adjacent to hour 0, just as 359° is adjacent to 1°), a sequential or cyclical color scale is the most appropriate choice for example, a gradient running from a light, cool color (blue) at hour 0 (midnight) through to a deep, warm color (dark red) at hour 23, as shown in the attached visualization. This lets the viewer trace the full daily wind pattern at a glance: points shift smoothly from cool to warm hues as the day progresses, so clusters of similar color reveal periods when the wind direction and speed were relatively stable, while a spread of colors across a similar region indicates that condition recurred at different times of day.

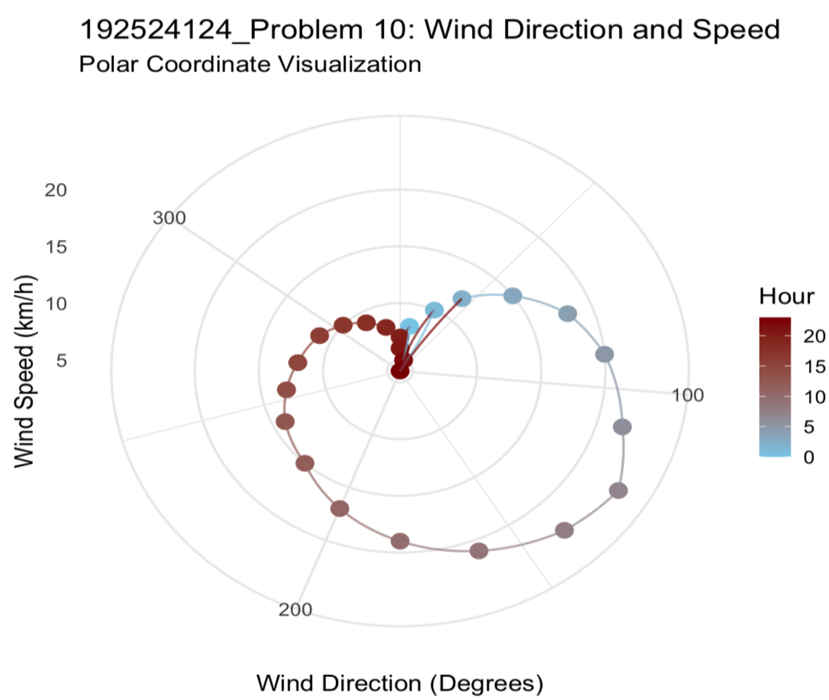

Figure 1: Polar Coordinate Visualization of Wind Direction and Speed by Hour

Problem 20 (10 Marks)

Scenario: An environmental agency wants to show carbon emissions change relative to a baseline year (some regions improved; others worsened) across 20 countries.

Chart Type	Choropleth (color-filled) world map
Coordinate System	Geographic (latitude, longitude)
Aesthetic Mapping	Fill color of each country = % carbon emissions change vs. baseline
Scale Type	Diverging numeric scale, centered at 0% (baseline)
Color Scale	Diverging Red White Blue

a) Design a color scheme appropriate for this "change relative to baseline" data. Justify the type of color scale chosen.

Design & Justification:

The appropriate design is a diverging color scale specifically for a Red White–Blue scheme applied to a choropleth map of the 20 countries. A diverging scale is justified because the underlying variable, percentage change in carbon emissions relative to a baseline year, has a meaningful, non-arbitrary reference point at 0% (no change), and values can move in two qualitatively different directions from that point: positive values (emissions increased, i.e., the situation worsened) and negative values (emissions decreased, i.e., the situation improved). In the proposed scheme, countries that worsened are shaded in increasingly saturated red as their percentage increase grows (e.g., up to +10% or more), countries that improved are shaded in increasingly saturated blue as their percentage decrease grows (e.g., down to −5% or beyond), and countries with negligible change sit in pale, near-white tones near the center of the scale. Grey is reserved separately for countries with no data available, so missing data is never confused with a true zero-change value. This scheme lets a viewer instantly separate the map into two visually distinct groups worsening (warm/red) and improving (cool/blue) regions while the saturation within each group still communicates the magnitude of change.

b) What color would you assign to a country with no change from the baseline, and why is this the correct choice?

Design & Justification:

A country with 0% change from the baseline should be assigned white (or a very pale neutral tone), placed exactly at the midpoint of the diverging scale. This is the correct choice because white is perceptually neutral; it carries no warm (red) or cool (blue) bias, so it does not lean the viewer's eye toward either the 'improved' or 'worsened' interpretation. Placing it precisely at the scale's center also gives the map a clear, immediately visible reference line: any hint of red or blue tinting on a country instantly signals that some change has occurred, and the direction of that tint (toward red or toward blue) tells the viewer, without needing to consult the legend, whether that change was for better or for worse.

c) Explain why a sequential color scale would misrepresent this particular dataset.

Design & Justification:

A sequential color scale (e.g., light yellow to dark green, or light to dark blue) encodes only the magnitude of a value along a single hue, running from low to high. It is designed for data that has a natural minimum and increases in one consistent direction, such as population density or rainfall totals where 'more' always means moving further along the same scale. Applied to this carbon-emissions-change dataset, a sequential scale would collapse two opposite and equally important facts a country that cut emissions by 5% and a country that increased emissions by 5% into the same visual language of 'how far from the low end of the scale,' with no way to distinguish which direction the change went. Depending on how the scale were oriented, either the improving countries or the worsening countries would end up looking visually 'better' (lighter or darker) purely as an artefact of the color ramp, actively misleading policymakers into misreading which countries genuinely improved and which worsened. Because the data has a true, meaningful zero point (the baseline) and two distinct directions of change around it, only a diverging scale not a sequential one can represent it accurately.

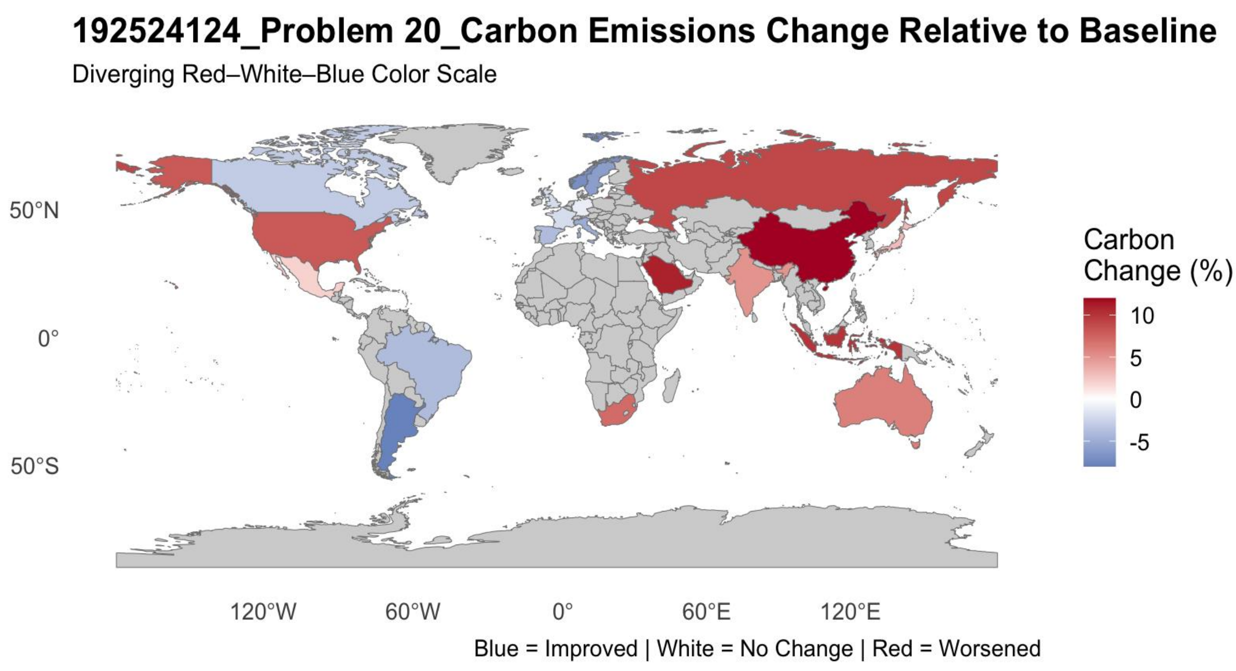

Figure 2: Diverging Red White–Blue Choropleth of Carbon Emissions Change by Country